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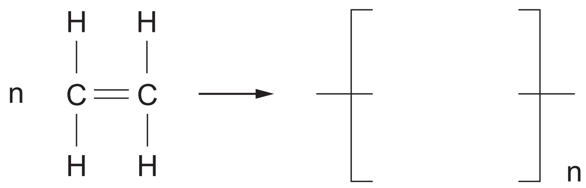
7. The first three members of the alkene family are shown below.

C_2H_4 C_3H_6 C_4H_8

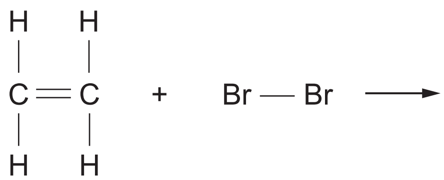
(a) (i) Give the **general** formula of the alkene family. [1]

....

(ii) I. Complete the equation for the polymerisation of ethene by drawing the structure of the repeating unit. [1]



II. Complete the equation by drawing the structure of the product formed when ethene reacts with bromine. [1]



III. Name the type of reaction that occurs in parts I and II above. [1]

....

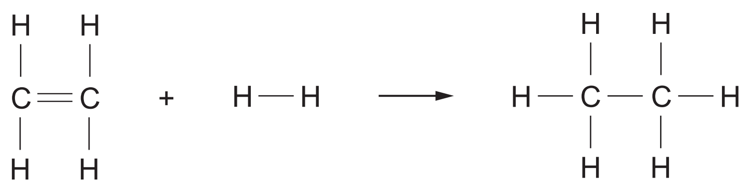
(iii) Ethene burns in air forming carbon dioxide and water.

Complete and balance the equation for this reaction. [2]

$C_2H_4 + 3O_2 \longrightarrow \boxed{}$ .... $+ \boxed{} H_2O$

(b) Ethene reacts with hydrogen, forming ethane.

The equation shows the bonds that are broken and the bonds that are formed in the production of ethane.



The relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
C—H	412
H—H	436
C═C	612

(i) Use the information in the table to show that the **total** energy needed to break all the bonds in the **reactants** is 2696 kJ. [2]

(ii) The **total** energy released when the bonds in the **products** are formed is 2820 kJ.

Calculate the energy released when forming a C—C bond. [2]

Energy = ... kJ

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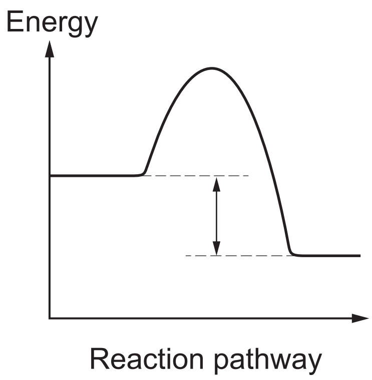
(iii) Use the total energy values given in parts (i) and (ii) to answer this question.

Tick (✓) the box next to the energy profile diagram that shows the reaction taking place. [1]



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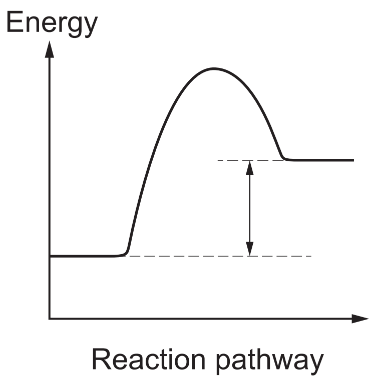

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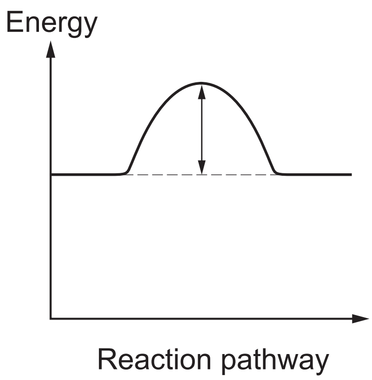

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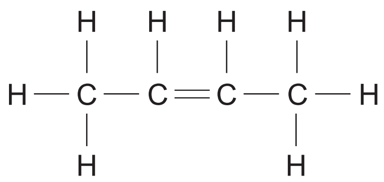
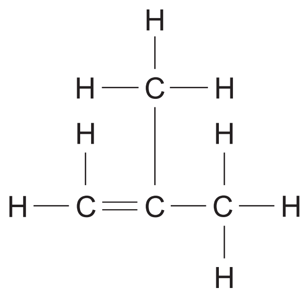
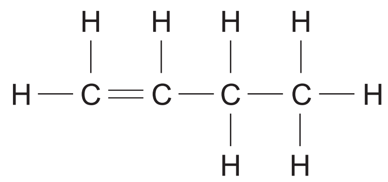
(c) Isomers **A**, **B** and **C** all have the molecular formula C_4H_8.



A



B



C

Give the **letter** of the structure fitting each of the following names. [2]

but-1-ene ....

but-2-ene ....

2-methylpropene ....

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